

Unit -2 E-Governance Models [12 Hrs.]

Model of Digital Governance

Models of digital governance are still evolving in developing countries. A few generic models have shaped up, which are finding greater recognition and are being replicated. These models are based on the inherent characteristics of ICT such as enabling equal access to information to anyone who is the part of the digital network and de---concentration of information across the entire digital network, connecting all sources of information. Information does not reside at any particular node in the Digital Governance Models but flows equally across all the nodes – a fundamental change from the more common hierarchical information flow model that leads to unequal distribution of information which skew power relations.

Hierarchy is inherent in the government departments. Equity based information flow may not be always compatible with government functioning. Therefore, appropriate administrative reforms and some reengineering may be required before e---governance may be really implemented. The models of governance are different in developing countries and developed countries due to difference in basic conditions and perspectives and expectations from good governance.

The six generic models of digital governance in developing countries are:

- Broadcasting/ Wider Dissemination Model,
- Critical Flow Model,
- Comparative Analysis Model,
- Mobilization and Lobbying Model,
- Interactive-service Model
- E-Governance Maturity Model

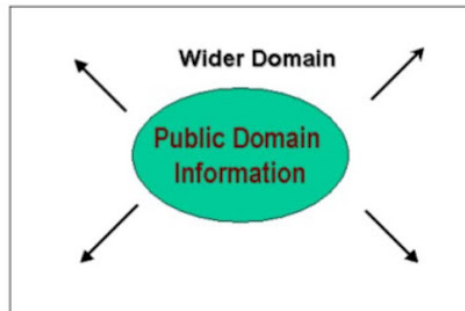
Broadcasting Model/ Wider Dissemination Model

Principle

- **Broadcasting model is based on the mass dissemination of governance-related information which is already available in the public domain into the wider public domain using ICTs.**
- This raises awareness among the citizens about ongoing governance processes and government services that are available to them and how they can benefit from them.
- The application of this model using appropriate technologies, could reduce the "information failure situations" where citizens are unaware of new and existing services being provided by the government.

Broadcasting / Wider Disseminating Model

Public Domain → Wider Public Domain



Applications:

- Putting governmental laws and legislations online
- Making available the names, contact addresses, emails, and mobile numbers of local/ regional/ national government officials online.
- Make available information such as governmental plans, budgets, expenditures, and performance reports online.
- Putting key judicial decisions of value to general citizens and creating precedence for future actions online. viz. key environmental decisions, state vs. citizen decisions, etc.

Some Organisations / Projects based on such models

- **India:** [Directory of Official Websites of Government of India](#)

National Informatics Centre (India) is the official website of the Government of India which makes available Ministerial Information, Indian Laws and Legislations Online, Contact Details etc. online for public access.

<http://goidirectory.nic.in/exe.htm#min> (Parent Website : <http://www.nic.in>)

- **Brazil:** [Official government website](#)

The website provides comprehensive information on Brazilian government as well as links to integrated citizen services.

<http://www.brazil.gov.br>

- **South Africa:** [The PIMS Monitor](#)

Idasa's Political Information and Monitoring Service (PIMS) providing an easy-to-use reference and record (with plain-language summaries) of all [bills](#), [acts](#) and [policy documents](#) that pass through parliament. The Monitor aims to help audience engage with democracy, intervene in the legislative process and make submissions to parliament in South Africa.

<http://www.pims.org.za/monitor/>

- **Global: [Earth Negotiations Bulletin](#)**

A reporting service which keeps informed citizens worldwide about global environmental negotiations, processes and decisions. It has a great value for citizens and even government officials in developing countries to keep track of global negotiations taking place in the West and be more informed about them.

<http://www.iisd.ca/voltoc.html> (Parent Website : <http://www.iisd.ca>)

- **Online Newspapers, Newsgroups and Portal Sites** - these are instrumental in bringing key political and other important information into the wider public domain.

Electorales.com - <http://www.electorales.com> A political Internet news and tools provider for Spanish-speaking markets around the world.

All Africa - <http://www.allafrica.com> Provides access to political news relating to various countries in Africa.

Chapter 2 Network - <http://www.advocacy.org.za> : The Chapter 2 Network is a clearinghouse of information and communication for social justice issues in South Africa. Through its website, it provides information about advocacy campaigns; training on Advocacy and lobbying; research on political intelligence, policy analysis and legislation monitoring and networking opportunities to interact with other civil society organisations.

Evaluation

- The Broadcasting Model is a stepping stone to more complex Digital Governance models. It is also the most fundamental model as it enhances the 'access' and 'flow' of information to a segment of society, which is essential to bringing good governance.
- In several developing countries, even the basic governance-related information is unavailable for instance, the opening and closing times of government offices, or information on government departments and units that deal with specific issues such as agricultural subsidies or getting a license for opening a business.
- Central and state governments in developing countries need to aggressively adopt this model to provide greater governance services to their constituencies, and to enhance the participation of citizens in the governance process.

Merits :

- **It enhances 'access' and 'flow' of information to all segments of the society.**
- Government can use this model to provide greater governance services to their constituencies and to enhance the participation of citizens in governance processes.

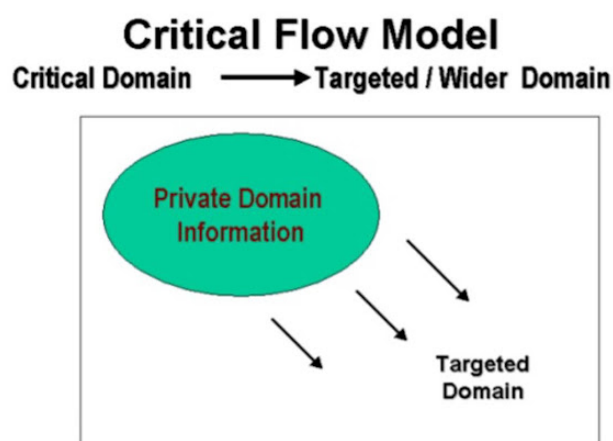
Demerits:

- **The model can lose its effectiveness**
 - In the situation of optimal ignorance
 - In societies, where the free flow of information is not possible.

Critical Flow Model

Principle

- The model is based on broadcasting information of 'critical' value (which by its very nature will not be disclosed by those involved with bad governance practices) to a targeted audience using ICTs and other tools.
- The targeted audience may include media, affected parties opposition parties, judicial bench, independent investigators, or the general public.
- Those who would divulge such information could include upright officials and workers, whistleblowers, affected parties, and those who were themselves involved in bad governance practices but have now changed their minds or may wish to trade such information for lenient punishments.



Application

This model could be applied in the following possible ways:

- Making available corruption-related data about a particular Ministry / Division/ Officials online to its electoral constituency or to the concerned regulatory body.
- Making available Research studies, Enquiry reports, Impact studies commissioned by the Government of Independent commissions to the affected parties.
- Making Human Rights Violations cases violations freely available to Judiciary, NGOs, and concerned citizens. Making available information that is usually suppressed, for instance,
- Environmental Information on radioactivity spills, effluents discharge, information on green ratings of the company to the concerned community.

Example of Critical flow model

Global: Wikileaks

www.wikileaks.org

WikiLeaks is a not-for-profit media organization. It brings important news and information to the public. It publishes secret information, news leaks, and classified media from anonymous sources to keep the public informed, amongst others on cases of corruption, whistle-blowing, and crimes.

Organisations / Projects based on such models

- **Global: [Transparency International -Daily Corruption News](#)** : A trial service is being run by Transparency International called the "The Daily Corruption News" which reports on corruption from around the world. The daily corruption news has been coming out since May 2000.
See http://www.transparency.org/press_moni.html for the daily updates.

- **Bangladesh : [Human Rights Portal](#)**

<http://www.banglarights.net>

The portal will actively promote human rights reforms both within Bangladesh and across geographical and political boundaries, and will support women, children, and marginalized communities in resisting social oppression. It will look at attempts by global forces (powerful governments, TNCs, international organizations) through surveillance, covert mechanisms and military and economic superiority to exploit and control smaller nations and communities.

Read the Significant Cases Section :

<http://www.banglarights.net/HTML/significantcases.htm>

- **India: [Central Vigilance Committee](#)**

<http://cvc.nic.in>

The website provides free-access information to citizens about government officials who have been indicted on judicial charges relating to corruption and have been advised penalty. People can also file complaints against any public servant who fall within the jurisdiction of the Commission.

See <http://cvc.nic.in> or go to:

- [Web-list of officers advised Major Penalty](#)
- [Procedure to lodge a complaint through e-mail](#)

- **China: [Human Rights in China](#)**

<http://iso.hrichina.org/iso/>

Human Rights in China (HRIC) is an international non-governmental organization founded by Chinese scientists and scholars in March 1989. It strives to make information on human rights issues accessible to the Chinese people through short-wave Chinese radio broadcasts, overseas Chinese publications, and an evolving virtual media hub on human rights.

See Human Rights New section:

<http://iso.hrichina.org/iso/news.adp>

- **El Salvador: Probidad**

<http://www.probidad.org/>

Probidad promotes democratization efforts vis-à-vis diverse and integrated anti-corruption initiatives, most which rely on the use of ICT and an extensive network of contacts. The activities are designed to monitor corruption and control mechanisms; mobilize awareness about the complexities and costs of corruption and increased interest and participation in curbing it; enhance the anti-corruption capacity of other civil society organizations, media, government, business, and researchers in our region; and contribute to more informed local and context-specific measures that undermine corruption and promote good governance.

- **India: Land/Property Registration in Andhra Pradesh**

Land is a major resource and a reason for conflict in areas where pressure on land is high. In developing countries, land related information such as ownership maps etc. is available only in the hands of few which increases the chance of its misuse. Land registration offices throughout Andhra Pradesh now operate computerized counters to help citizens to complete registration requirements within an hour instead of several days, as was necessary under the earlier system. The lack of transparency in property valuation under the old system resulted in a flourishing business of brokers and middlemen leading to corruption.

<http://www1.worldbank.org/publicsector/egov/cardcs.htm>

Evaluation

- **This model is more directed and evolved in comparison to the Broadcasting / Wider-Dissemination Model.** Different organizations can use it differently depending on the aspect of governance and the situation they want to address.
- **Critical Flow Model is more focused in terms of its information content and its intended users.** Due to the critical aspect of information, **the model exposes the weakest aspects of governance and decision-making mechanisms** and informs people about specific cases of state failure and bad governance to build up a case for concerted action.
- **The model will not work in cases where government mechanisms do not foster public debates and censure all information of critical nature.**
- **It will also fail where the government maintains tight control of overall information.** There it remains restricted to the top few levels of the government. Inherently the Internet is an open medium.
- **Thus, restricted dissemination is only typical-only those interested may use the critical and subject-based information lodged on Internet websites for public access (as indicated above) as applications.**

Merits

- This model is more directed and evolved.
- Different organizations can use it differently depending on the aspect of governance the situation they want to address.
- The model corrects information failure, raising awareness about the bad governance practices, and acts as a hindrance to bad governance practices.
- Also, the model exerts indirect pressure on the concerned governance institution/policy-making body to move away from optimal ignorance attitude to reform, and take into cognizance the interest and opinion of the masses in decision-making processes.

Demerits

- The model may not work in cases where the governance mechanism does not allow public debates and opinions and censures all information of critical nature.

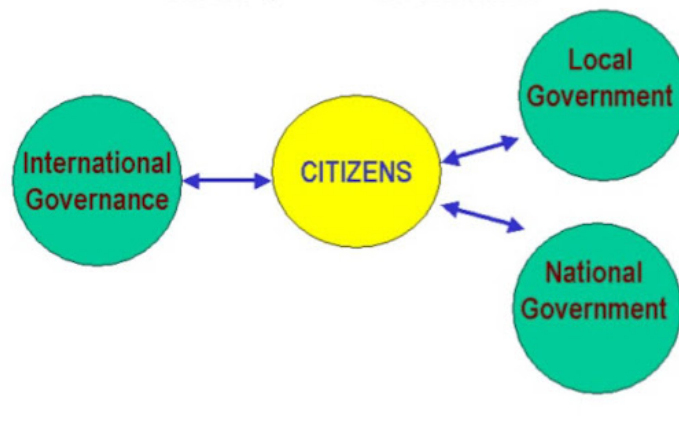
Interactive Service Models

Principle

- **It is a consolidation of the other digital governance models and opens up possibilities for one-to-one and self-serviced participation of individuals in governance processes. (or opens up for direct participation of individuals in the governance process.)**
- The Interactive- service model makes possible various services offered by the government to **be directly accessible to citizens.**
- **the participation is direct and not through representatives.**
- **It can bring greater objectivity and transparency in decision-making processes, and give a greater feeling of involvement and empowerment, provided that individuals are willing to engage in the governance processes.**
- Under this model, the various services offered by the government become directly available to its citizens by opening up an **interactive Government to Consumer to Government (G2C2G) channel** in various aspects of governance.

Service Delivery Model

Citizen $\longleftrightarrow$ Government



Applications

This model could be applied in the following possible ways:

- To establish an interactive communication channel with key policy-makers and members of planning commissions.
- To conduct electronic ballots for the election of government officials and other office-bearers.
- To conduct public debates/opinion polls on issues of wider concern before the formulation of policies and legislative frameworks.
- Filing of grievances, feedback, and reports by citizens with the concerned governmental body. Establishing decentralized forms of governance.
- Performing governance functions online such as revenue collection, filing of taxes, governmental procurement, payment transfer, etc.

Organisations / Projects based on such models

<http://www.gyandoot.net/gyandoot/intranet.html>

India: Gyandoot Gyandoot is an intranet in the Dhar district connecting rural cybercafes catering to the everyday needs of the masses. The site has the following services to offer in addition to the hope that it has generated by networking, the first district in the state of Madhya Pradesh in India

- Commodity/ Agricultural Marketing Information System
 - Copies of land maps
 - On-Line Registration of Applications
 - Public Grievance Redressal
-
- **Philippine Customs Reform** : Using an "off-the-shelf" customs application package as the main building block, the Philippines Customs Bureau has developed an on-line system to process clearance of imports, payment of duty, and delivery of release orders for shipments to leave the docks. The new on-line system has lessened the cost of trade for

businesses, reduced opportunities for fraud, and helped the Bureau to maximize revenue collection.

<http://www1.worldbank.org/publicsector/egov/philippinecustomscs.htm>

- **Central Vigilance Committee (India)** - pioneering initiative toward e-vigilance. It provides free-access information to citizens about government officials who have been indicted on judicial charges relating to corruption and have been advised penalty. People can also file complaints against any public servant who fall within the jurisdiction of the Commission.

<http://cvc.nic.in>

- [Procedure to lodge a complaint through e-mail](#)
- [Web-list of officers advised Major Penalty](#)

Evaluation

- This model is more embedded in digital-governance initiatives in the developed countries and has often been proposed for implementation in developing countries. Such forms of solution transfer may not be very effective.
- **The model is on the higher end of technology reliance as compared to the other models. This makes it difficult to replicate in developing countries in the absence of individual and secure ICT access.**
- **Various other issues also need to be considered carefully before such blind duplication can be attempted in developing countries.**
- **However, the trend is definitely in this direction and sooner or later will be implemented in all the countries with due modifications for local adaptation.**

Comparative Analysis Model

Principle

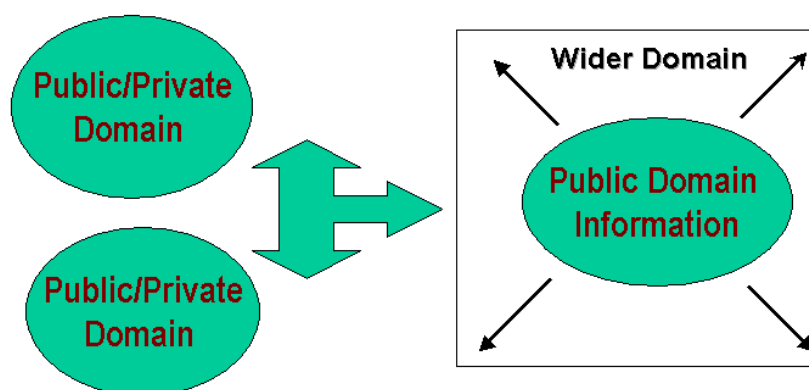
- Comparative Knowledge Model is one of the least-used but a highly significant model for developing country which is now gradually gaining acceptance. **The model can be used for empowering people by matching cases of bad governance with those of good governance, and then analyzing the different aspects of bad governance and its impact on the people.**
- The model is based on using ICT to explore information available in the public or private domain and comparing it with the known information sets. **The outcome is strategic learnings and arguments**, for instance, if a given amount of money can build '5' schools in village 'A' then why does the same amount of money build only '2' schools in village 'B'?
- Essentially, the model continuously assimilates Best Practices in the areas of governance and then uses them as benchmark to evaluate other governance practices. It then uses the result to advocate positive changes or to influence 'public'

opinion on these governance practices. The comparison could be made over a time scale to get a snapshot of the past and present situation or could be used to compare the effectiveness of an intervention by comparing two similar situations.

- The strength of this model lies in the infinite capacity of digital networks to store varied information and retrieve and transmit it instantly across all geographical and hierarchal barriers.

Comparative Analysis Model

Private / Public Domain + Public / Private Domain
→ Wider Public Domain



Applications/ Possible Projects

This model could be applied in the following possible ways:

- To learn from past policies and actions and derive learning lessons for future policy-making.
- To evaluate the effectiveness of the current policies and identify key learnings in terms of strengths and flaws in the policies.
- To effectively establish conditions of Precedence, especially in the case of Judicial or legal decision-making (example for resolving patent-related disputes, public goods ownership rights), and use it to influence/ advocate future decision-making.
- To enable informed decision-making at all levels by enhancing the background knowledge and also providing a rationale for action.
- To evaluate the performance and track-record of a particular decision-maker/ decision-making body.

Organisations / Projects based on such models

- **Global: Human Development Indicators** - The Human Development Report of UNDP makes use of archived Statistical information pertaining to literacy, health, national income etc. as a benchmark to assess the progress made by different countries with regards to their Human Development Index and suggests policy recommendations based on that.

<http://hdr.undp.org/reports/global/2002/en/indicator/indicator.cfm?File=index.html>

Many countries now also prepare national level indicators to compare progress made in different states in a country.

http://hdr.undp.org/reports/view_reports.cfm?type=3

- **India: [Comparative Learning from Disasters](#)** : In the wake of earthquake in Kutch in India (January 2001), there was a lot of comparative learning relating to disaster management drawn from a high intensity earthquake which shook Latur in India in 1993. The extensive information available on internet on both these earthquakes open up vast scope of comparison by all segments of the society.

<http://www.cddc.vt.edu/digitalgov/Latur-Gujarat.htm>

(a presentation by Mr.Praveen Singh Pardesi- Indian Administrative Service (IAS) officer of the Maharashtra cadre and currently Joint Secretary to the Chief Minister of Maharashtra, India) Comparison of different Disasters can make people realise that damages to life and property incurred are not just a factor of intensity of the disaster but also dependent on the preparedness of the Government machinery and conducive government policies to handle that disaster.

- **Philippines : [Vote.ph](#)** is the non-partisan, non-sectoral online directory and quick reference center on Philippine elected government officials and election candidates for both the national and the local levels. As such, it has a dual purpose of helping the voters know who are running in their respective area and who their incumbent elected officials are.

<http://www.vote.ph>

- **[Green Rating of Ministers \(India\)](#)** : The Centre for Science and Environment in India in 1999 started an online and offline survey of how Green its Chief Ministers are. The results of it are published in print and partially on the website to enable public to keep track of environmental performance of their Chief Ministers.

http://www.cseindia.org/html/au/au4_022099.htm

- **[France : Opening of a Platform for Comparison of programs of Electoral Candidates](#)** - The site compare according to 15 topics the principal proposals of the electoral candidates for the town hall of Paris. The platform also presents the electoral system of the capital, the biographies of the candidates to the town hall of Paris as well as a list with accompanying notes of links relating to the local elections. In addition to information which it will offer to the Parisian voters, this experimental project has as an ambition to open a discussion on the interest for the democratic life of comparative electoral sites. Its users are invited to make share of their comments and reactions for the development of the site for the next elections.

<http://www.sciences-po.fr/etudiants/electionsM/index.htm>

- **[Friends of the Earth Political Action Committee \(FoEPAC\)](#)**: The Friends of the Earth PAC is a non-partisan Political Action Committee supporting candidates with good environmental records and defeating candidates with poor environmental records.

<http://www.foepac.org/>

Discussions

- **Developing countries could very effectively use this comparative model as ICT opens their access to the global and local knowledge products at a relatively low -cost.** The model is very much based on the existing sets of information but requires the ability to analyse and bring out strong arguments which could then be used to catalyze existing efforts towards self governance.
- **There is a vast scope of application of this model for Judicial advocacy as Landmark/Key Judgments of the past could be used as precedence for influencing future decision-making.** Further, **watch-guard organizations and monitor-groups can use this model to continuously track the governance past record and performance and compare with different information sets.**
- **The model however becomes ineffective in absence of a strong civil society interest and public memory which is essential to force decision-makers to improve existing governance practices.**

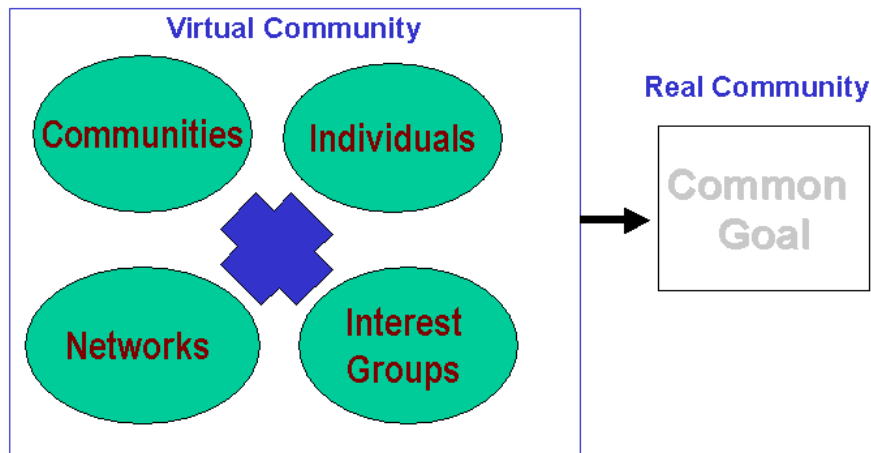
E-Advocacy / Mobilization and Lobbying Model

Principle

- **E-Advocacy / Mobilization and Lobbying Model is one of the most frequently used Digital Governance model and has often come to the aid of the global civil society to impact on global decision-making processes.**
- **The model is based on setting-up a planned, directed flow of information to build strong virtual allies to complement actions in the real world.**
- **Virtual communities are formed which share similar values and concerns, and these communities in turn link up with or support real-life groups/ activities for concerted action. The model builds the momentum of real-world processes by adding the opinions and concerns expressed by virtual communities.**
- **The strength of this model is in its diversity of the virtual community, and the ideas, expertise and resources accumulated through this virtual form of networking.** The model is able to mobilize and leverage human resources and information beyond geographical, institutional and bureaucratic barriers, and use it for concerted action.

Mobilisation and Lobbying Model

Networking Networks for Concerted Action



Applications

This model could be applied in the following possible ways:

- **Fostering public debates on issue of larger concerns, namely on the themes of upcoming conferences, treaties etc.**
- **Formation of pressure groups on key issues to force decision-makers to take their concerns into cognisance.**
- Making available opinions of a suppressed groups who are not involved in the decision-making process into wider public domain.
- **Catalysing wider participation in decision-making processes.**
- Building up global expertise on a particular theme in absence of localised information to aid decision-making.

Organisations / Projects based on such models

- **Global: [Greenpeace Cyber-activist Community](http://cybercentre.greenpeace.org/t/s/community_articles)** - an effort towards creation of virtual communities to mobilise global support against some of the disputable environmental policies/ actions of the Government. Since the cyberactivist system began in June 2000, there are now 116794.0 registered cyberactivists, who participated in 357003 action alerts and sent 160597 e-cards to individuals and organizations.
http://cybercentre.greenpeace.org/t/s/community_articles
- **Global: [Drop the Debt Campaign](http://www.jubileeusa.org/)** - the campaign spreads awareness of their activities through emails and mobilises support of concerned individuals, and encourages them to directly express their concern to key decision-makers (by making available their email and other contact addresses).
<http://www.jubileeusa.org/>

- **Global: [Independent Media Centre](http://www.indymedia.org/)** - The Center was established by various independent and alternative media organizations and activists for the purpose of providing grassroots coverage of the World Trade Organization (WTO) protests in Seattle in 1999. The center acted as a clearinghouse of information for journalists, and provided up-to-the-minute reports, photos, audio and video footage through its website.
<http://www.indymedia.org/>
- **Global: [IGC Internet](http://www.igc.org)** - Institute for Global Communications (IGC) has changed the way the progressive community worked by introducing them to email, online discussions, mailing lists and the Web. IGC Internet offers progressive individuals and groups a place on the Internet to learn, meet and organize. At the moment, IGC Internet is changing to take its community of activists and non-profit professionals into the Internet's next wave by focusing on content, information sharing and new, collaborative tools.
<http://www.igc.org>

Discussions

This model has grown manifold since the onset of debates on the Seattle round of World Trade Organisation in 1999, which saw the formation of several virtual communities to express their concerns in the WTO agreements. The display of a unified, informed civil society force at Seattle was in some ways a result of the intensive interaction and exchange of opinion happening over the virtual networks months prior to this WTO summit. There was a lot of concerted actions at the Regional level as an end result of such discussions which built into the global movement.

The model enhances the scope of participation of individuals and communities in debates which affect them and help them build a global alliance. A community may no longer find itself isolated but may find an ally for mobilizing effective action through this model. It also creates an effective deterrent for governments and decision-making bodies who are responsive to people's opinion to provide better governance.

The model could also be used favourably by the government in a positive manner to encourage public debates on issues where the opinion and expertise of civil society is of great importance and therefore could become a tool to enhance democratic practises and improve governance practices (especially in Developing Countries).

E-Governance Maturity Model

Evolution

The E-Governance Maturity Model (EMM – 1.0), based on the conventional software maturity models, proposes some levels of maturity, depending on the effectiveness with which the e-governance efforts have been initiated, implemented or successfully complete.

The model also provides for identification of key focus areas that need to be concentrated for attaining a specific maturity level.

	Level 1: Simple Website	Level 2: Online Government	Level 3: Integrated Government	Level 4: Transformed Government
Attributes	Static pages Lists of departments and contact information Links to separate departments Policy statements Downloadable forms and documents Access primarily via telephone No site reporting, tracking or analysis	Departmental focus Online forms for applications and registrations Online payment Request information or service via email Respond to online surveys Limited online help, FAQs, resolution services Basic account inquiry Basic benefits enrollment	End-to-end electronic transactions Automated RFP and procurement process Cross-departmental sharing of information Automated advice and problem resolution data Limited configuration capabilities Self-service HR administration Web-based training	Community-centric, integrated, intergovernmental processes Common platform for targeting content through any channel/touch point Internal/external business process integration and collaboration (planning, workflow, design) Constituent case tracking to ensure resolution and satisfaction Highly configurable HR (benefits, career planning, development training)

Traditionally, e-governance has been defined as an ICT enabled route to good governance with a view to enhancing transparency in the system and provide prompt and quality services to the citizen. E---governance is an evolutionary path, whose effective implementation requires a complete understanding of its various elements and at the same time taking a holistic view to stay focused on its overall objectives.

E-Governance journey encounters several milestones that need to be identified and modeled so that efforts invested can be assessed and an appropriate course of action are taken by the organization to further its way on the path of e---governance. The E---governance Maturity Model (EMM – 1.0) proposes 5 levels of maturity, depending upon the effectiveness with which the e--- governance efforts have been initiated, pursued, utilized and institutionalized. EMM 1.0 will facilitate government organizations to assess the current level of e---governance initiatives and accordingly make efforts for the future. The model further identifies the characteristics exhibited by organizations at various levels of maturity that will facilitate correct assessment of the current status. The model also provides Key Focus Areas that need to be focused on to attain a particular maturity level.

Five Maturity Levels

The E-governance Maturity Model (EMM—version 1.0) is based on the fact that speed, openness and ubiquity are some of the major capabilities of ICTs, which can be leveraged for generating transparency, responsiveness and accountability in the system, on the one hand and empowering the common man by providing faster access to right information at the right time, on the other. It is also based on a service-oriented approach, where public administration is seen as a professional activity and efficient delivery of services to the internal and external users (customers) is emphasized as a key performance indicator of the government department. The internal customers/users of an organization are its employees and the external users are the citizens, businesses, other government and non-governmental bodies that the organization needs to satisfy in its e-governance endeavour. However, there may exist organizations that are performing well even without ICT application to its functions. Even such organizations may gradually take up to ICT for the sake of other reasons and other benefits.

The maturity levels, described below, provide a necessary mechanism to benchmark the efforts invested by an organization in implementing e-governance and subsequently sustaining it to the satisfaction of its customers/users.

Level 1: Closed

This is the stage when an organization does not use ICT as a facilitator for good governance and has no plans to do so in the near future. This situation may arise due to lack of exposure to ICTs and associated benefits that again may depend upon a number of reasons; remoteness from the mainstream in terms of location is primary, and lack of resources and strategic thinking could be some of the other issues. As a result, the organization is 'closed' in terms of being connected and sharing of information in the context of "e"-governance. However, even in this condition the organization may be efficiently functioning. Given the trend today, all organizations may take up ICT implementation sooner or later.

Level 2: Initial

This level corresponds to the stage when an organization has initiated the automation of its processes but on an ad-hoc basis. No organized efforts are made to undertake the e-governance initiatives. Also, due to lack of direction many such e-governance efforts are abandoned at a subsequent date.

Level 3: Planned

The e-governance initiative, at this level, is undertaken with a systematic approach. The organization has a clearly defined vision, objectives and goals for e-governance. A need assessment study is conducted to prioritize areas of implementation and gauge the extent of e-readiness. Taking input from the need assessment study, extensive planning has been carried out indicating policies, strategies, various activities, stakeholders, roles and responsibilities and resources required in terms of time, money and manpower to undertake the e-governance exercise. However, the organization is yet to enter into the planned implementation of the e-governance exercise, even though all the requisite planning is completed.

Level 4: Realized

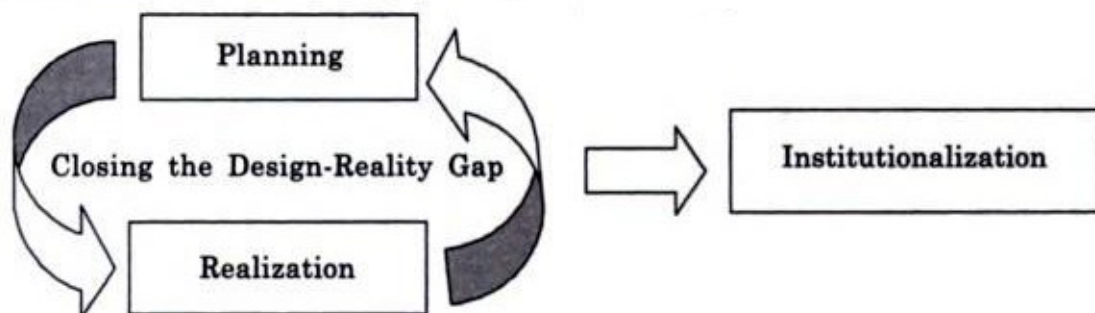
This level corresponds to the stage when the organization actually realizes the complete e-governance plan. Consequently, an integrated system is established where all the internal processes of the organization are computerized and there is a seamless information exchange among all concerned entities. The organization starts delivering the services to its external as well as internal customers/users in an effective manner.

Complete realization of the plan, in a single instance, would entail enormous amount of resources in terms of time, money and manpower, which may necessitate adopting a phased approach for operationalizing the e-governance services. Accordingly, a further classification within this level has been proposed that measures the extent of realization of the plan over a period of time. These sub-levels are also indicative of the openness and effectiveness with which the information is exchanged among the various entities of the organization (external and internal).

- (a) **Retrospected.** At this level, the organization has retrospectively studied its business processes in view of its vision, overall e-governance objectives, the service-oriented approach (wherein government is expected to effectively deliver the services to its customers/users), and changes, if required, in the processes are initiated as a constant evolutionary process.
- (b) **E-ready.** In this stage, e-readiness essentials,* which are also the building blocks for e-governance, are ensured by the organization.
- (c) **Partially open.** At this stage some of the e-governance services are operationalized resulting in a partial information exchange among the entities, both within and outside the organization.
- (d) **Open.** This sub-level of realized state implies complete deployment of e-governance services that ensure an integrated system that is open to information exchange. The focus here shifts from acquiring and implementing “e” enabling factors to effectiveness with which the services are delivered. The system gains responsiveness to deal with the customer needs and is accountable for its services.

Level 5: Institutionalized

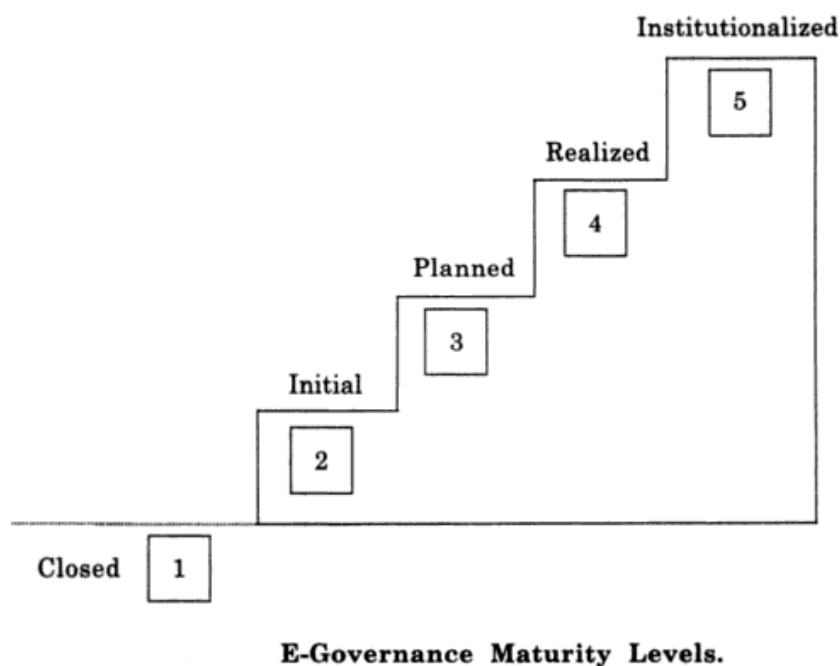
At this level, the organizations sustain the realized state over a period of time so that e-governance becomes part of its work culture. The e-governance services are effectively utilized and accepted by the users. Several iterations between planned and realized state lead to institutionalization, when e-governance becomes a way of life



Reaching the Institutionalized Stage.

Following define the e-readiness of an organization:

- (a) **Presence of strategic thinking, leadership and commitment among top-level decision makers.** It is necessary to have visionaries at the top level who can put e-governance on the agenda of the organization and are able to carry it through to implementation and effective utilization.
- (b) **Institutional infrastructure.** There should also be institutions responsible for creating e-governance awareness among stakeholders and users, and coordinate the e-governance exercise.
- (c) **ICT infrastructure.** A sound computing and communication infrastructure is an essential requirement for effective operationalization of e-governance services.
- (d) **Human capacities.** It is important to build human capacities in terms of necessary knowledge and skills to initiate, implement and sustain e-governance initiatives. It is equally important to generate right attitude that is receptive to ICT based administration and ICT based delivery of services. All require extensive and intensive training and ICT orientation programmes at various levels of the staff and officers in the concerned government organization.
- (e) **Legal infrastructure.** The necessary laws and regulations should also be in place to support ICT as a tool for good governance.



Characteristics of Maturity Levels

The characteristics exhibited by the organizations at various levels of maturity facilitate assessing the current level of e-governance attained by the organization. The following points describe the characteristics of the organizations at various levels:

Level 1: Closed

- **Organizations are closed to e-governance. No plans or vision is available. They continue with fully manual and conventional operations.**

Level 2: Initial

- **The organization lacks strategic thinking and direction for e-governance at top level.**
- **There are unorganized and isolated efforts of automation in some areas.**
- **Automation efforts are a result of individual's initiatives.**
- **The effort sustains as long as the enthusiasm of the individual, who initiated it, remains and is often abandoned due to lack of direction.**
- **Such organizations generally accumulate hardware without any planning and much of it goes unutilized or underutilized.**

Level 3: Planned

- **E-governance is a part of an organization's agenda.**
- **Strategic thinking and leadership guide the e-governance initiatives. A clear understanding of e-governance needs as projected by the external and internal customers/users.**
- **An extensive plan is prepared for implementing e-governance, addressing all Key Focus Areas (KFAs) and other related issues.**
- **All the necessary documents for e-governance planning are in place. These documents include Vision and Scope document for e-governance, Need Assessment Survey document, Policy guidelines, and Action Plan and Outsourcing guidelines**

Level 4: Realized

a) Retrospected

- **Business processes are attuned to the vision and overall e-governance objectives. There is awareness about e-governance among all concerned-the stakeholders as well as the customers/users.**

b) E-ready

- **The organization has a sound infrastructure (technological, institutional, legal, and human) in place, for implementing e-governance. The customers/users are oriented and motivated to use e-governance services.**

c) Partially open

- **Some of the e-governance services are deployed, leading to partial information exchange among the entities.**
- **Partially open organizations sometimes focus only on their internal or backend processes, allowing an information exchange within the confines of the organization**

thus remaining insulated from its external entities. In such cases, Government-to-Employee (G2E) interface is visible, whereas Government-to-Citizen (G2C), Government-to-Government (G2G), and Government-to-Business (G2B) interfaces are not yet established.

- Another case of partial deployment of e-governance services results in a conspicuous G2C interface with no emphasis on building G2E, G2B, or G2E interfaces. In an enthusiasm to quickly open up its front-end, the organization negligibly focuses on computerization of the supporting backend processes, thus creating hollowness behind the face of static websites.
- A hybrid of the above two cases is characterized by building interfaces with external entities and simultaneously focusing on computerizing the corresponding backend processes, thus opening a balanced information exchange among the internal as well as the external customers of the organization.

d) Open

- **The organization has an integrated system, reflective of smooth information exchange within and outside the organization, i.e. Government-to-Employee (G2E), Government-to-Citizen (G2C), Government-to-Government (G2G) and Government-to-Business (G2B) interfaces are well established over a sound foundation of e-governance building blocks (the e-readiness essentials).**
- The organization focuses on satisfying the users of e-governance services. The internal and external customers/users of the organization start utilizing the e-governance services and become dependent on them.

Level 5: Institutionalized

- **The e-governance system of the organization is driven by a well-established Knowledge Management System that generates an ability in the organization to evolve with time in view of new requirements.**
- E-governance becomes an effortless exercise for the organization and it becomes a way of life for the stakeholders and customers/users. The organization at this level is completely paperless.

Key Focus Areas

Key Focus Areas (KFAs) indicate the areas that need to be focused by an organization and are a set of related activities when performed collectively, help achieving a particular level of maturity. KFAs are defined from level 3 (Planned) onwards, as this is the stage from where e-governance effort is systematically attempted.

Planned

- Define a quantifiable vision for the e-governance exercise.
- Conduct a Needs Assessment Survey in view of objectives covering the following areas:
 - Survey of requirements both within and outside the organization, indicative of the information needs of the internal (employees) and external (citizens, business and other government agencies) customers/users with respect to e-governance.
 - Analyze the requirements to identify priority areas for initiating e-governance exercise.
 - Assess the extent of e-readiness for identified areas and requirements to achieve the desired level of e-readiness.
- Prepare an extensive plan on e-governance that includes the following areas:
 - Define objectives and goals for the e-governance exercise.
 - Design policies and strategies for implementing e-governance.
 - Decompose the e-governance exercise into various activities.
 - Project the resource requirements in terms of time, money and manpower.
 - Identify stakeholders and assign roles and responsibilities.
 - Define implementation methodology.
 - Define measures for creating awareness and change in mindset among the external and internal customers/users of the organizations for effective implementation of e-governance.
 - Define measures for attaining required level of e-readiness.
 - Identify risk factors and propose risk mitigation plan.
 - Define the expected impact and propose an impact assessment methodology.
 - Identify external sources of funds, if required.
 - Define mechanisms (Research and Development, Knowledge Management initiatives) for developing innovative ways of delivering services within the organization and outside it.
 - Define the time for which the plan is valid.
 - Prepare all the necessary documentation including Vision and Scope document for e-governance, Need Assessment Survey document, Policy guidelines, Action Plan and Outsourcing guidelines.

Realized

- Arrange for resources required to implement the e-governance initiative.
 - Develop a high level awareness and commitment among decision-makers, stakeholders and users to initiate and carry forward the e-governance objectives.
 - Select vendors for outsourced activities and formalize terms and conditions with all the stakeholders, clearly assigning roles, responsibilities and ownership.
 - Set up management committees with appropriate representation of all stakeholders for executing and monitoring the e-governance exercise.
 - Conduct a detailed study and review of the existing business processes in view of e-governance objectives.
 - Initiate the change in business processes wherever required, and bring in suitable legislation to make it effective.
 - Acquire and/or design, develop, test and deploy e-governance services with the following issues in mind:
 - Address information needs of common man in local language with easy to use interface.
-
- Address the issues related to standardization of content and data to facilitate seamless flow of information among concerned entities.
 - Ensure interoperability in terms of interconnectivity, data integration and information access.
 - Use open standards.
 - Ensure easy accessibility to information.
 - Provide efficient data communication.
 - Build scalable architecture.
 - Ensure wide market support (supply from multiple vendors).
 - Ensure wide product support (interconnection between products from diverse vendors).
 - Ensure cost effectiveness.
- Conduct extensive training for customers/users and administrators for effective operationalization and utilization of e-governance services.
 - Collect, compile, validate and update data/content.
 - Maintenance activities.
 - Conduct an Impact Analysis to assess the effectiveness of service delivery.
 - Initiate activities (surveys, research and development, knowledge management initiatives) to devise more innovative ways of developing and delivering e-governance services.

Institutionalized

- Address the *design-reality* gaps, if any, by iterating between planning and realization phases.
- Evolve a mechanism (knowledge management system, research and development initiatives and surveys) to make e-governance an effortless exercise so that the entire system develops an ability to evolve and scale up with time and new requirements.

Towards Good Governance through E-Governance Models.

The digital governance models bring about a transformation in the existing forms of governance as they change the nature of citizen-governance relationship and bring in new agents and mechanisms to influence the governance processes. They foster democratic control over the governments' economic, social and welfare policies by citizens and civil society organizations—a key process requirement for good and responsive governance.

The changes brought about in the citizen-governance relationship through digital governance are fourfold:

1. They open up avenues for flow of information both vertically and laterally to encompass a wider foundation of the civil society. A greater density of information flow is achieved between government and civil society, or within civil society itself. The right to voice and expression gets gradually embedded among citizens through digital means.
2. Information becomes difficult to be capitalized by a few for political gains at the expense of ignorance of citizens. The power equations shift from being concentrated at select nodes to its even distribution among citizens, opposition parties and watchdog groups.
3. There is a greater scope to influence policy-makers and members of civil society through collective opinion, direct participation, participation in public debates and use of advocacy tools.
4. Policy-makers become more aware of the voices of people and can effectively involve them in policy-making mechanisms. They realize that their actions are under the scrutiny of many more watchdog organizations and there are greater avenues available with people to obtain any information. Information also becomes difficult to obliterate and is forever archived to increase the institutional memory of society.

It may, however, be noted that all these features described are applicable only to well established democracies with sound systems in place. This criterion may not be applicable to many developing countries and even some developed countries.

The widespread application of digital governance models synergizes representative forms of democracy with direct participatory forms. This becomes possible because information earlier residing with citizen's representatives in the governance domain is now available with the citizens themselves. People

are therefore more aware of the political issues which interests them, and also about the implication of the actions made by their representatives.

This form of informational egalitarianism creates an effective watchdog system where people watch those who are supposed to guard their interests in the governance mechanisms. This can potentially catalyse popular democracy at local, regional and national levels.

People's Participation in ICT-Enabled Governance

	<i>Conventional media</i>	<i>ICT and convergence media</i>
Mode of participation	Representative of <i>ex-situ</i>	Individual/Collective <i>in-situ</i>
Form of participation	Passive/Reactive	Pro-Active/Interactive
Impact of participation	Indirect	Direct/Immediate

As seen from the comparison chart, the involvement of people in ICT-enabled governance mechanisms is significantly different in comparison to traditional forms of governance. The mode of governance transforms from "representative" to "individual based" and from "passive" to "proactive". People are no longer totally dependent on information provided to them by their leaders or policy-makers through conventional media to form an opinion about the issue. They are able to capitalize on different sources of information and can proactively enter into a dialogue with the decision-makers on issues of interest to them.

Since ICT-enabled governance models can directly connect individuals with officials in the government and decision-makers, the impact is immediate, and puts greater access and control over governance mechanism in the hands of individuals. Spaces therefore get created within the existing governance mechanisms that would be democratically informed by citizen's voice. This process makes the government more responsible and accountable to its citizens.

Digital governance models bring with them the potential to bring in far-reaching changes in governance structures in developing countries. The approach to it however has to be cautious and well thoughtout. Simple replication of popular e-governance models of the developed world in the developing world will not be effective and may end up marginalizing people who are on the wrong side of the digital divide. Local needs and priorities should guide the selection and nature of implementation of e-governance models in an evolutionary manner.

Creativity lies in the use of information and blending it with the potential offered by ICT to create customized low-cost digital governance models. Solutions may often emerge by analysis of reasons behind governance failures by the people themselves and then imaginatively using digital governance models to focus on the weakest spot to bring gradual improvements in governance. Small successes are significant landmarks in the path of transformation to good governance.